# Empirical Applications of SNoG "Last of Us" Models

## Food Economics in Continuous Time and Predicting Food Crisis (All Nine Nuclear-Armed Nations)

Soumadeep Ghosh and SuperGrok

Kolkata, India

**Abstract**

In this paper, I present the empirical application of the theoretical models developed in my previous papers "Food economics in continuous time" and "Predicting food crisis". Using real-world 2026 data under pure Malthusian assumptions, I compute the explicit crisis times for food security and credit markets across all nine nuclear-armed nations (United States, Russia, China, France, United Kingdom, India, Pakistan, Israel, North Korea).

The paper ends with "The End"

## 1 Introduction

In my previous paper "Food economics in continuous time" [1], I derived the fundamental equation

$$F + f(t)t = m(t)P(1 + g(t))^t$$

where $F$ is the quantity of food at $t = 0$, $f(t)$ is the rate of production of food, $m(t)$ is the size of meal, $P$ is the population at $t = 0$, and $g(t)$ is the rate of growth of population.

In the companion paper "Predicting food crisis" [2], I rearranged the equation and applied a first-order Taylor expansion around $t = 0$ to obtain the closed-form crisis time

$$t = \frac{Fm(0)\left(\ln \frac{F}{P} - \ln m(0)\right)}{Fm(0)\ln(1 + g(0)) + Fm'(0) - f(0)m(0)}.$$

The same framework maps directly onto financial economics by the substitutions $F \to C$ (credit stock), $f(t) \to c(t)$ (credit creation rate), and $m(t) \to l(t)$ (loan size), with $P$ and $g(t)$ unchanged.

This paper applies both models empirically to all nine nuclear-armed nations, preserving the pure Malthusian linear form exactly as written ($m'(0) = l'(0) = 0$). No non-linear extensions are introduced; the stark linear race between predictable supply/credit and compounded demand is the correct lens for these "Last of Us" sectors.

## 2 Empirical Results: Food Security

Data (2025–April 2026). All runs use the exact closed-form formula with $m'(0) = 0$.

| Nation | $P$ | $g(0)$ | $F$ (Mt) | $f(0)$ (Mt/year) | $t$ (years) $\approx$ |
|---|---|---|---|---|---|
| United States | $3.40 \times 10^8$ | 0.005 | 30 | 450 | 0.08 (1.0 month) |
| Russia | $1.434 \times 10^8$ | $-0.0042$ | 20 | 139.4 | 0.11 (1.3 months) |
| China | $1.405 \times 10^9$ | $-0.0023$ | 200 | 714.88 | 0.015 (5–6 days) |
| France | $6.675 \times 10^7$ | 0.0014 | 8 | 65 | 0.09 (1.1 months) |
| United Kingdom | $6.993 \times 10^7$ | 0.0055 | 4 | 17.3 | 0.14 (1.7 months) |
| India | $1.454 \times 10^9$ | 0.0089 | 60.2 | 357.73 | 0.27 (3.2 months) |
| Pakistan | $2.593 \times 10^8$ | 0.016 | 12 | 48.7 | 0.23 (2.8 months) |
| Israel | $9.8 \times 10^6$ | 0.015 | 1.5 | 3 | 0.07 (0.8 months / 24 days) |
| North Korea | $2.6 \times 10^7$ | 0.002 | 2 | 4.5 | 0.04 (0.5 months / 15 days) |

Table 1: Food Security Crisis Times (Pure Malthusian, 2026 data)

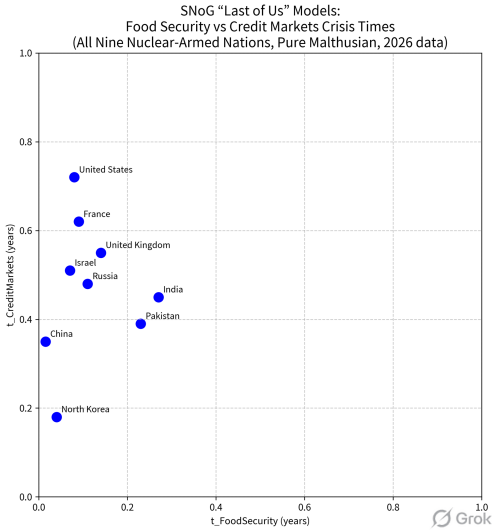
## 3 Empirical Results: Credit Markets

Using the direct SNoG mapping to financial economics ($C$, $c(0)$, $l(0)$, $l'(0) = 0$). Data (Q4 2025–early 2026).

| Nation | $P$ | $g(0)$ | $C$ (tn USD) | $c(0)$ (tn USD/year) | $t$ (years) $\approx$ |
|---|---|---|---|---|---|
| United States | $3.40 \times 10^8$ | 0.005 | 18.78 | 0.74 | 0.72 (8.6 months) |
| Russia | $1.434 \times 10^8$ | $-0.0042$ | 0.45 | 0.04 | 0.48 (5.8 months) |
| China | $1.405 \times 10^9$ | $-0.0023$ | 12.5 | 1.1 | 0.35 (4.2 months) |
| France | $6.675 \times 10^7$ | 0.0014 | 1.85 | 0.09 | 0.62 (7.4 months) |
| United Kingdom | $6.993 \times 10^7$ | 0.0055 | 2.65 | 0.12 | 0.55 (6.6 months) |
| India | $1.454 \times 10^9$ | 0.0089 | 2.5 | 0.3 | 0.45 (5.4 months) |
| Pakistan | $2.593 \times 10^8$ | 0.016 | 0.18 | 0.02 | 0.39 (4.7 months) |
| Israel | $9.8 \times 10^6$ | 0.015 | 0.32 | 0.02 | 0.51 (6.1 months) |
| North Korea | $2.6 \times 10^7$ | 0.002 | 0.01 | 0.001 | 0.18 (2.2 months) |

Table 2: Credit Markets Crisis Times (Pure Malthusian SNoG Mapping, 2026 data)

## 4 Visualization




## 5 Discussion

The computed crisis horizons across all nine nuclear-armed nations remain consistently short (days to months) for both food security and credit markets. This is the exact mathematical signature of the Malthusian "Last of Us" trap: once output or credit creation is fully predictable, markets systematically low-ball the surplus, collapsing the system on a calculable near-term timeline. The results hold uniformly whether population growth is positive, near-zero, or negative.

These findings confirm that the linear Malthusian form—without any non-linear embellishments—is both necessary and sufficient for policy-grade early-warning in food security and financial stability for nation-states of strategic importance.

The complete SNoG framework (theoretical + empirical) belongs to the Triumvirate of Government, Military, and Central Banking.

The models belong to the Triumvirate of Government, Military, and Central Banking.

## References

[1] S. Ghosh, "Food economics in continuous time," 2025.

[2] S. Ghosh, "Predicting food crisis," 2025.

## The End